

MOOSE 1D Cahn-Hilliard Discussion

A cleaned-up transcript and working notes from the conversation: setting up a 1D Cahn-Hilliard simulation in the MOOSE framework, fixing input-file issues, and visualizing/exporting results in ParaView.

1. Goal: run a 1D Cahn-Hilliard simulation

User: Can you help me do a 1d Cahn-Hilliard simulation in the moose framework?

The suggested approach was to use the split Cahn-Hilliard formulation, solving for concentration c and chemical potential w . The essential equations are:

$$\begin{aligned}dc/dt &= \text{div}(M \text{ grad } w) \\ w &= dF/dc - \kappa_c \text{ laplacian}(c) \\ F(c) &= c^2 (1-c)^2\end{aligned}$$

A minimal 1D spinodal-decomposition input file was proposed, with a generated 1D mesh, variables c and w , random initial concentration near 0.5, SplitCH kernels, a parsed double-well free energy, and Exodus output.

```
[Mesh]
  type = GeneratedMesh
  dim = 1
  nx = 400
  xmin = 0
  xmax = 200
[]

[Variables]
  [c]
    order = FIRST
    family = LAGRANGE
  []
  [w]
    order = FIRST
    family = LAGRANGE
  []
[]

[ICs]
  [c_ic]
    type = RandomIC
    variable = c
    min = 0.48
    max = 0.52
    seed = 12345
  []
  [w_ic]
    type = ConstantIC
    variable = w
    value = 0
  []
[]

[Kernels]
  [c_dot]
    type = CoupledTimeDerivative
    variable = w
    v = c
  []
  [w_res]
    type = SplitCHWRes
    variable = w
    mob_name = M
```

```

[]
[c_res]
  type = SplitCHParsed
  variable = c
  f_name = F
  kappa_name = kappa_c
  w = w
[]
[]

[Materials]
[free_energy]
  type = DerivativeParsedMaterial
  property_name = F
  coupled_variables = c
  expression = 'c^2*(1-c)^2'
  derivative_order = 2
[]
[constants]
  type = GenericConstantMaterial
  prop_names = 'M kappa_c'
  prop_values = '1.0 1.0'
[]
[]

```

2. MOOSE checkout versus application directory

User: Are these the files I should see in a "moose checkout"? [listing from ~/moose/projects/myfirst]

The directory shown was identified as a generated MOOSE-based application, not the top-level MOOSE framework checkout. The executable myfirst-opt belongs to that application. A typical layout was described as:

```

~/moose/
framework/
modules/
scripts/
test/
tutorials/
projects/
  myfirst/
    myfirst-opt
    Makefile
    src/
    include/
    test/

```

The input file can live inside the application directory and be run with `./myfirst-opt -i ch_1d.i`, provided the application includes the `phase_field` module. Otherwise, use the phase-field executable or a phase-field-enabled app.

3. Periodic BC block fix

User: MOOSE error: missing required parameter BCs/periodic/type

The problem was the nesting and capitalization of the periodic boundary condition block. The lowercase `[periodic]` subblock was interpreted as a normal BC object. The corrected block uses `[BCs][Periodic]`.

```

[BCs]
  [Periodic]
    [c_periodic]
      variable = c
      primary = left
      secondary = right

```

```

        translation = '200 0 0'
    []
    [w_periodic]
        variable = w
        primary = left
        secondary = right
        translation = '200 0 0'
    []
[]

```

If the boundary names left/right are not accepted, numeric IDs 0 and 1 may work for a 1D GeneratedMesh:

```

[BCs]
  [Periodic]
    [c_periodic]
      variable = c
      primary = 0
      secondary = 1
      translation = '200 0 0'
    []
    [w_periodic]
      variable = w
      primary = 0
      secondary = 1
      translation = '200 0 0'
    []
  []
[]

```

4. End-time convergence message

User: The run reached time = 100, then failed on a tiny extra timestep around $dt = 1e-12$.

The key point was that the simulation had essentially succeeded: it reached Time Step 1000, time = 100, and converged. The later failure came from an extra floating-point endpoint/cleanup step.

```

Time Step 1000, time = 100, dt = 0.1
Solve Converged!

```

Two suggested fixes were to add a timestep tolerance, if supported, or replace end_time with num_steps to avoid floating-point endpoint ambiguity.

```

[Executioner]
  type = Transient
  solve_type = NEWTON

  dt = 0.1
  end_time = 100
  timestep_tolerance = 1e-10

  nl_abs_tol = 1e-10
  nl_rel_tol = 1e-8
  l_tol = 1e-5
  l_max_its = 50
  nl_max_its = 20
[]

[Executioner]
  type = Transient
  solve_type = NEWTON

  dt = 0.1
  num_steps = 1000

  nl_abs_tol = 1e-10

```

```

nl_rel_tol = 1e-8
l_tol = 1e-5
l_max_its = 50
nl_max_its = 20
[]

```

5. ParaView setup for 1D results

User: What ParaView settings should I use? Properties. Display. View.

For a 1D Exodus result, the best visualization is usually concentration c versus x using Plot Over Line. The original Exodus object should be applied, variables c and w loaded, and the display colored by c . Then use Plot Over Line from $x=0$ to $x=200$.

```

Properties:
  Point Variables: c, w
  Apply

Display:
  Representation: Surface
  Coloring: c

Filter:
  Filters -> Data Analysis -> Plot Over Line
  Point1: 0, 0, 0
  Point2: 200, 0, 0
  Resolution: 400

Chart:
  X Array Name: arc_length
  Y Series: c

```

User: Where do I set X Array: arc_length?

Select the PlotOverLine item in the Pipeline Browser, not the original Exodus file. Then use the Display section, often with advanced settings enabled using the gear icon. In many ParaView versions, Plot Over Line uses `arc_length` automatically; if the horizontal axis runs from 0 to 200, it is already correct.

User: I do not see "animate over time"

ParaView usually does not have a menu item literally named Animate over time. The animation controls are the VCR-style buttons at the top. If they are hidden, enable these toolbars:

```

View -> Toolbars -> VCR Controls
View -> Toolbars -> Current Time Controls

```

Use the play triangle or step buttons to move through saved timesteps. If only one timestep appears, the output file probably contains only one saved frame.

6. Output frequency and movie export

User: Can I export the movie?

Yes. In ParaView, once the animation plays correctly, use File -> Save Animation... and choose MP4 or AVI if available. A reliable cluster workflow is to export numbered PNG frames and convert them to MP4 with `ffmpeg`.

```

File -> Save Animation...

Suggested settings:
  Frame Rate: 10 or 15 fps
  Resolution: 1280 x 720
  Frame Window: all timesteps

```

```
ffmpeg -framerate 15 -i ch_1d_frame.%04d.png -pix_fmt yuv420p ch_1d_cahn_hilliard.mp4
```

User: interval = 10 under [Outputs] did not work

The corrected MOOSE output syntax for this case is to use an explicit Exodus output subblock and time_step_interval, not interval at the top level of [Outputs].

```
[Outputs]
[exodus]
  type = Exodus
  time_step_interval = 10
  execute_on = 'initial timestep_end final'
[]
[]
```

If CSV output is also wanted:

```
[Outputs]
[exodus]
  type = Exodus
  time_step_interval = 10
  execute_on = 'initial timestep_end final'
[]

[csv]
  type = CSV
  time_step_interval = 10
  execute_on = 'initial timestep_end final'
[]
[]
```

Before rerunning, delete stale output so ParaView opens the new file:

```
rm -f ch_1d_out.e ch_1d_out.e-s* ch_1d_out.csv
./phase-opt -i ch_1d.i
```

7. Compact current recommended input-file fragments

For the current working state, the most important corrected fragments are:

```
[BCs]
[Periodic]
[c_periodic]
  variable = c
  primary = left
  secondary = right
  translation = '200 0 0'
[]
[w_periodic]
  variable = w
  primary = left
  secondary = right
  translation = '200 0 0'
[]
[]

[Executioner]
  type = Transient
  solve_type = NEWTON
  dt = 0.1
  num_steps = 1000
  nl_abs_tol = 1e-10
  nl_rel_tol = 1e-8
  l_tol = 1e-5
  l_max_its = 50
  nl_max_its = 20
```

```
[]  
[Outputs]  
  [exodus]  
    type = Exodus  
    time_step_interval = 10  
    execute_on = 'initial timestep_end final'  
  []  
[]
```

End of discussion notes.